



Smart Crop Advisory

A Smart IoT and Web-based Solution
for Better Farming Decisions



Team Members

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“Smart decisions. Smarter harvests.”



Problem Statement



- Farmers face uncertainty due to unpredictable weather and agricultural variations that directly affect crop yield and income.
- Farmers often irrigate without knowing exact soil condition → wastage of water.
- Need : To build a Smart Crop Advisory System that uses sensors and data analytics to give real-time, weather-based recommendations to farmers.

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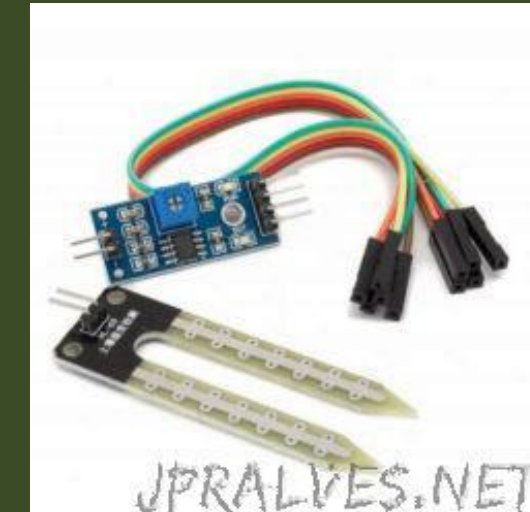
Our solution

- ❖ **Collect data from multiple sensors (temperature, humidity, moisture, water level, rain).**
- ❖ **Analyze , data using AI & weather models.**
- ❖ **Send real-time advisory alerts (irrigation, rainfall warning, soil dryness, etc.).**
- ❖ **Display all readings & advice on a web dashboard.**



Components Used

ESP32 Microcontroller
Soil Moisture Sensor
DS18B20 (waterproof temp sensor)
DHT22 (air temp +humidity)
Water pump +Relay
Breadboard, jumper wires, battery



Sensors collect data → microcontroller processes it → sends to server → displayed on web dashboard.

Working Model & Output

01

Sensors collect data in real-time (soil, temp, humidity, rain, tank level).

02

Data sent to cloud/server via Wi-Fi.

03

Website displays live readings and alerts.

Advisory suggestions generated (e.g., “Irrigation needed”, “Rain expected, avoid watering”).

DEBUG THUGS



Future Scope & Conclusion

- Add automation (pump ON/OFF based on threshold).
- Add fertilizer/pH sensors.
- Add AI-based irrigation scheduling.
- Use solar power for complete sustainability.

Conclusion:

“Empowering farmers through smart technology for a better tomorrow.”

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Thank You.



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